MODULE *BlockDag*

In this specification we define notions on *DAGs* useful for DAG-based consensus protocols (which build *DAGs* of blocks)

EXTENDS *Digraph*, *FiniteSets*, *Sequences*, *Integers*

CONSTANTS
  N The set of all nodes
, R The set of rounds
, $Leader(\_)$ operator mapping each round to its leader

$Max(S) \triangleq$ CHOOSE $x \in S : \forall\, y \in S : y \leq x$
$Min(S) \triangleq$ CHOOSE $x \in S : \forall\, y \in S : x \leq y$

For our purpose of checking safety and liveness, *DAG* vertices just consist of a node and a round.

$V \triangleq N \times R$ the set of possible *DAG* vertices
$Node(v) \triangleq v[1]$
$Round(v) \triangleq$ IF $v = \langle\rangle$ THEN 0 ELSE $v[2]$ accomodates $\langle\rangle$ as default value

Next we define how we order *DAG* vertices when we commit a leader vertice

$LeaderVertex(r) \triangleq$ IF $r > 0$ THEN $\langle Leader(r),\, r\rangle$ ELSE $\langle\rangle$
$IsLeader(v) \triangleq LeaderVertex(Round(v)) = v$
$Genesis \triangleq \langle\rangle$

OrderSet(S) arbitrarily order the members of the set S. Note that, in TLA+, CHOOSE is deterministic but arbitrary choice, *i.e.* CHOOSE $e \in S$: TRUE is always the same e if S is the same

RECURSIVE $OrderSet(\_)$
$OrderSet(S) \triangleq$ IF $S = \{\}$ THEN $\langle\rangle$ ELSE
  LET $e \triangleq$ CHOOSE $e \in S$: TRUE
  IN $Append(OrderSet(S \setminus \{e\}),\, e)$

$PreviousLeader(dag,\, r) \triangleq$
  IF $\exists\, l \in Vertices(dag) : IsLeader(l) \land Round(l) < r$
  THEN
    CHOOSE $l \in Vertices(dag)$:
      $\land\ IsLeader(l) \land Round(l) < r$
      $\land\ \forall\, v \in Vertices(dag) : IsLeader(v) \land Round(v) < r \Rightarrow Round(v) \leq Round(l)$
  ELSE
    $\langle\rangle$

RECURSIVE $Linearize(\_,\, \_)$
$Linearize(dag,\, l) \triangleq$ IF $Vertices(dag) = \{\langle\rangle\}$ THEN $\langle\rangle$ ELSE
  LET $dagOfL \triangleq SubDag(dag,\, \{l\})$
    $prevL \triangleq PreviousLeader(dagOfL,\, Round(l))$
    $dagOfPrev \triangleq SubDag(dag,\, \{prevL\})$

$$remaining \triangleq Vertices(dagOfL) \setminus Vertices(dagOfPrev)$$

IN $Linearize(dagOfPrev,\ prevL) \circ OrderSet(remaining \setminus \{l\}) \circ \langle l \rangle$

technically, we should use a topological sort instead of $OrderSet(\_)$